MLP for Heart Disease Prediction

Project Overview

This project implements and compares two Multi-Layer Perceptron (MLP) neural network architectures to predict the presence of heart disease using patient health data. The models are developed in Python using Scikit-learn and executed in Google Colab.

Objectives

- Load and preprocess the Heart Disease dataset.
- Split the dataset into training and testing sets.
- Build two MLP architectures with different hidden-layer configurations.
- Train and evaluate both models.
- Compare model performance using Accuracy, Precision, Recall, and F1 Score.
- Visualize the performance comparison.

Dataset

- **Source:** UCI Heart Disease Dataset
- **Records after preprocessing:** 297
- **Input features:** 13
- **Target:** Binary classification
    - 0 – No heart disease
    - 1 – Heart disease present
- **Missing values:** Removed during preprocessing.

Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

Model Architectures

Model 1: Simple MLP

- One hidden layer
- 16 neurons
- ReLU activation function

Model 2: Deep MLP

- Two hidden layers
- 64 neurons in the first hidden layer
- 32 neurons in the second hidden layer
- ReLU activation function

Both models use the Adam optimizer and are evaluated on the same test dataset.

Methodology

1. Load the Heart Disease dataset.
2. Handle missing values and convert the target into binary classes.
3. Split the data into 80% training and 20% testing sets.
4. Standardize input features using StandardScaler.
5. Train both MLP architectures.
6. Generate predictions on the test set.
7. Calculate Accuracy, Precision, Recall, and F1 Score.
8. Visualize and compare the evaluation metrics.

Evaluation Metrics

- **Accuracy:** Measures the proportion of correct predictions.
- **Precision:** Measures the proportion of positive predictions that are correct.
- **Recall:** Measures how many actual positive cases are correctly identified.
- **F1 Score:** Combines precision and recall into a single metric.

Results

The two MLP architectures are compared using a performance table and a bar chart. The better-performing architecture is selected based on its actual evaluation results.

Add your final metric values and comparison graph after running the notebook.

How to Run

1. Open Google Colab.
2. Upload or open the project notebook.
3. Run the code cells in sequence.
4. Review the evaluation metrics and generated graph.

Conclusion

This project demonstrates how neural network architectures with different hidden-layer configurations can affect heart disease prediction performance. Comparing the evaluation metrics helps identify a suitable architecture for the dataset.

Disclaimer: This project is intended for educational purposes only and is not a clinically validated medical diagnostic tool.

Author

Developed as part of a machine learning assignment.